

Evanns Morales-Cuadrado

Robotics PhD Candidate | Georgia Institute of Technology | U.S. Citizen | Fulbright Specialist

CONTACT **Email:** evannsmc@gmail.com **GitHub:** <https://github.com/evannsmc>
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SUMMARY Robotics PhD candidate in Dr. Samuel Coogan's FACTS Lab at Georgia Tech and AI for Autonomy intern at Sandia National Laboratories, specializing in safe autonomy, reinforcement learning, and real-time control. Proven track record of developing and deploying safe, computationally efficient planning and control methods for autonomous systems, with work published in leading peer-reviewed robotics and controls venues. Selected to the U.S. Department of State's Fulbright Specialist Roster after spearheading an international research collaboration with a Brazilian partner institution.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native), Portuguese (B1)

EDUCATION

Georgia Institute of Technology 2022-Present
Ph.D. in Robotics
Doctoral Minor in Optimization Methods

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington 2018-2022
Honors Bachelor of Science in Electrical Engineering
Minor in Mathematics
Minor in Physics
Certificate in Unmanned Vehicle Systems

- GPA: 3.9/4.0 Summa Cum Laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

WORK EXPERIENCE

Year-Round Graduate Research Intern | AI for Autonomy May 2026 - Present
Sandia National Laboratories Albuquerque, NM

- Applied reinforcement learning to autonomy problems in support of national security
- Designed and deployed robotics and automation infrastructure that enables multi-robot experiments
- Built C#/.NET networking frameworks and execution scripts that coordinate communication and task execution across multi-robot systems, now used team-wide

Graduate Research Assistant | FACTS Lab August 2022 - Present
Georgia Institute of Technology Atlanta, GA

- Advised by Dr. Samuel Coogan
- Developed a lightweight tracking control law that beat PX4's native tracker in aggressive quadrotor flight and matched NMPC's tracking accuracy at far lower computation time and CPU energy
- Developed safe planning and control methods that combine reachability analysis, formal methods, optimization, and learning to generate fast, provably safe trajectories under unknown disturbances
- Built the lab's real-time flight-test stack, enabling repeatable hardware validation

- Teaching Assistant** | Vertically Integrated Project in Robotics January 2024 - May 2026
Georgia Institute of Technology Atlanta, GA
- Mentored about 20 undergraduates per semester on hands-on robotics research, with multiple mentees going on to robotics PhD programs
 - Designed and delivered undergraduate lectures on ROS, Git/GitHub, Linux, and control theory, preparing students for research on hardware
- Research Assistant** | Autonomous Systems Lab November 2021 - May 2022
University of Texas at Arlington Arlington, TX
- Advised by Dr. Frank Lewis
 - Contributed to reinforcement-learning quadrotor control research under Dr. Frank Lewis, then ranked 12th in the USA and 23rd worldwide
 - Completed UTA Honors College thesis on lightweight Newton-Raphson tracking control for computationally constrained aerial systems
- Teaching Assistant** | Graduate-Level Intelligent Systems Course January 2022 - May 2022
University of Texas at Arlington Arlington, TX
- Selected as an undergraduate to TA a graduate-level course
 - Led exam review sessions and graded homework, exams, and projects
- Research Assistant** | Dynamical Networks and Control Lab October 2019 - May 2022
University of Texas at Arlington Arlington, TX
- Advised by Dr. Yan Wan
 - Conducted original research on learning-based minimum-time and minimum-energy multi-vehicle path planning using ROS and OpenCV

SELECTED PUBLICATIONS

- E. Morales-Cuadrado**, S. Coogan, Z. Kakish, and A. Parikh. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA)*. (title redacted until the double-blind review process is complete)
- L. Pinto*, **E. Morales-Cuadrado***, S. Coogan, M. A. Wehrmeister, L. Caselatto, and C. E. Magrin. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA)*. *Equal contribution. (title redacted until the double-blind review process is complete)
- M. Cao, A. Harapanahalli, **E. Morales-Cuadrado**, and S. Coogan, “Trajectory Tracking for Systems with Unknown Time-Varying Disturbances.” *IEEE Open Journal of Control Systems (OJ-CSYS)*, 2026.
- E. Morales-Cuadrado**, “Computationally Efficient and Safe Control for Aerial Robotic Systems under Threat and Disturbance.” Ph.D. Thesis Proposal, Georgia Institute of Technology, 2026. [pdf available](#). [slides available](#).
- E. Morales-Cuadrado**, L. Chung, S. Kousik, and S. Coogan, “RTD-RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances.” Joint Submission to 2026 Modeling, Estimation and Control Conference (MECC) and ASME Letters in Dynamic Systems and Control (ALDSC). (accepted to both) [preprint available](#).
- E. Morales-Cuadrado**, L. Baird, Y. Wardi, and S. Coogan, “Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method.” *IEEE Transactions on Control Systems Technology (TCST)*, 2026. [preprint available](#).
- L. Baird, **E. Morales-Cuadrado**, and S. Coogan, “Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic.” Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (under revision) [preprint available](#).
- E. Morales-Cuadrado**, C. Llanes, Y. Wardi, and S. Coogan, “Newton-Raphson Flow for Aggressive Quadrotor Tracking Control.” *2024 American Control Conference (ACC)*. [preprint available](#).
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RESEARCH INTERESTS	• Safe Autonomy • Hardware Deployment • Unmanned Ground and Aerial Vehicles	• Advanced Nonlinear Control • Learning-Based Control • Trajectory Generation and Planning
OPEN-SOURCE CONTRIBUTIONS	• PX4 Autopilot: Contributed documentation improvements to the official PX4 Autopilot manual. • Vicon4PX4: Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech’s Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.	
SERVICE & LEADERSHIP	Co-Founder and President of Puerto Rican Student Association August 2024 - Present Georgia Institute of Technology Atlanta, GA	
	• Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture • Formed a three-member co-founder board and identified a faculty advisor • Raised funds for the organization and recruited members from campus • Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech	
	Undergraduate Research Mentor August 2021 - May 2022 University of Texas at Arlington Arlington, TX	
	• As the lead undergraduate student in Dr. Frank Lewis’ Autonomous Systems Lab, mentored junior undergrads in controls research and guided them on a research project in ground vehicle control • Guided students in Robot Operating System, Python, Linux, and Control Systems	
	Teaching Assistant Graduate-Level Intelligent Systems Course January 2022 - May 2022 University of Texas at Arlington Arlington, TX	
	• Selected as an undergraduate to TA a graduate-level course • Led exam review sessions and graded homework, exams, and projects	
HONORS AND AWARDS	Eta Kappa Nu Public Relations Chair & Circuits I Tutor August 2020 - May 2022 University of Texas at Arlington Arlington, TX	
	• Executive board member of UT Arlington’s IEEE-HKN chapter • In charge of social media advertising for the honor society’s events and outreach initiatives for the electrical engineering student body • Volunteered to provide free tutoring as a service to electrical engineering students in Circuits I	
	Fulbright Specialist August 2026 Peer-review-selected to the U.S. Department of State’s Fulbright Specialist Roster for a three-year term	
	Top-3 Finalist at a Deep Learning Research Symposium November 2023 Professor-sponsored award for novel research in deep learning and robotics	
	Goizueta Fellowship at Georgia Tech August 2022 Highly selective fellowship for graduate students of Latin-American descent	
	Summa Cum Laude Honors at University of Texas at Arlington May 2022 Exceptional academic distinction in top GPA tier for graduation honors	
	Honors College Degree Requirements at University of Texas at Arlington December 2021 Completed research projects in disparate fields throughout my undergraduate career with various professors and defended a senior capstone research project before the Honors College committee	
	Electrical Engineering Honors Scholar at University of Texas at Arlington December 2021 Inaugural member of the electrical engineering honors cohort	
	IEEE-HKN International Electrical Engineering Honor Society Membership August 2021 Selective undergraduate society recognizing outstanding academic achievement and community service	
	Research Experience for Undergraduates Fellowship Recipient August 2020 Fellowship to sponsor promising young undergraduate researchers	

Chance Vought Engineering and Science Endowment Scholarship	August 2020
Highly selective award given annually to three recipients across the entire College of Engineering	
Toray Composite Materials America Inc. Sons and Daughters Scholarship	May 2019
Academic scholarship awarded to children of employees of Toray America	
James E. Casey Scholarship Recipient	January 2019
Academic scholarship awarded to children of employees of UPS	
James R. Hoffa Memorial Scholarship Recipient	May 2018
Academic scholarship awarded to children of Teamsters Union Members	
Grapevine, TX Rotary Club Scholarship Recipient	May 2018
Local rotary club scholarship for academic and civically-minded scholars	
National Hispanic Scholar	March 2018
Recognition of top national performers of Latin descent on the SAT	
National Merit Scholar	March 2018
Recognition of top national performers on the SAT	